

IP Address & DHCP

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1. IP Versions

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2. IP Addressing

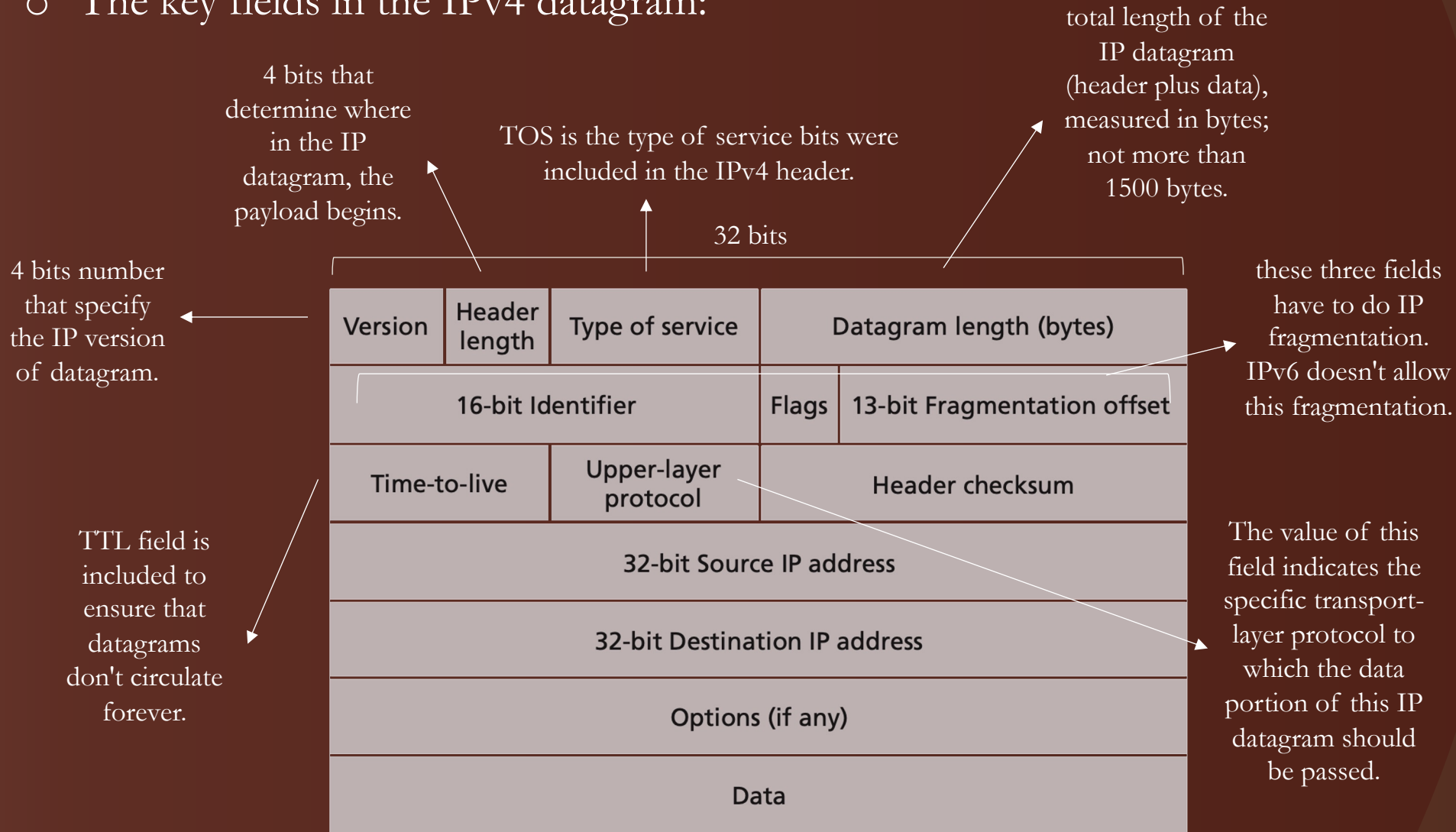
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3. DHCP

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1.1 IPv4 Datagram format:

- The key fields in the IPv4 datagram:

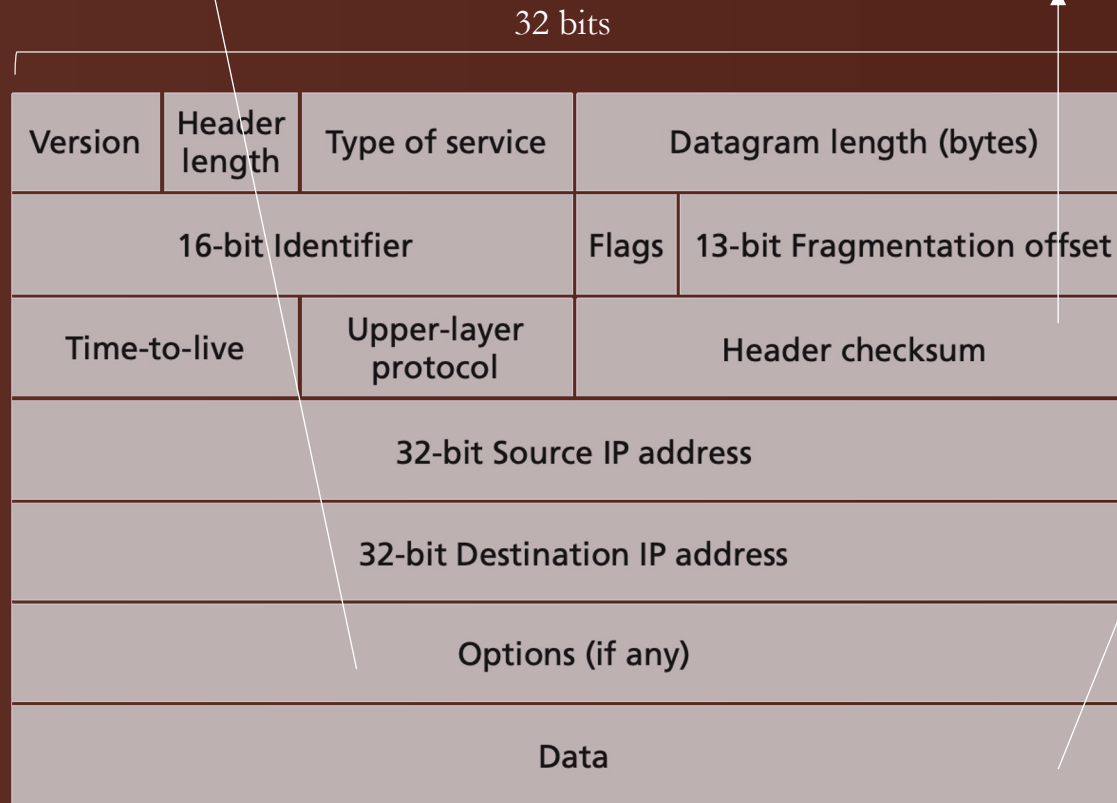


1.1 IPv4 Datagram format:

- The key fields in the IPv4 datagram:

The options fields allow an IP header to be extended. IP options were not included in the IPv6 header.

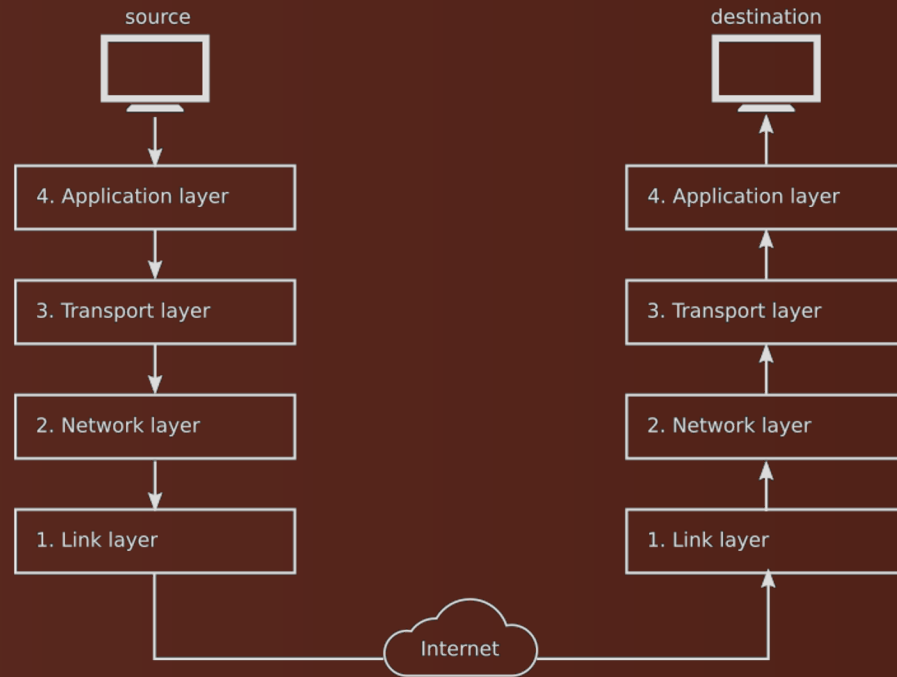
Internet checksum aids a router in detecting bit errors in a received IP datagram.



1.2 A Question?

- A question often asked at this point is, why does TCP/IP perform error checking at both the transport and network layers?

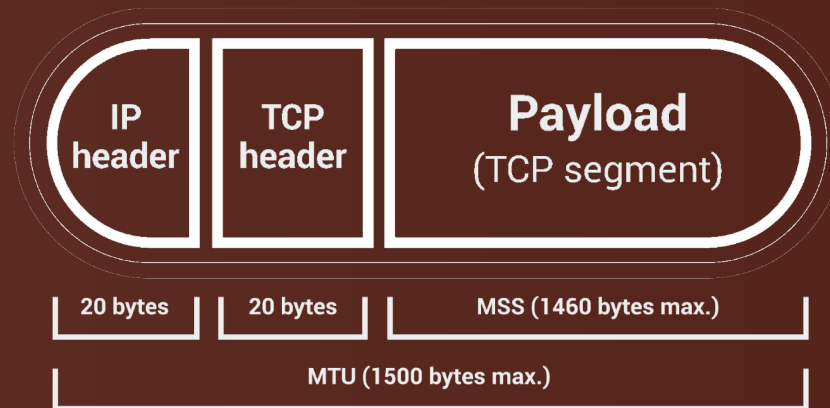
There are several reasons for this repetition. First, note that only the IP header is checksummed at the IP layer, while the TCP/UDP checksum is computed over the entire TCP/UDP segment. Second, TCP/UDP and IP do not necessarily both have to belong to the same protocol stack. TCP can, in principle, run over a different network-layer protocol (for example, ATM) and IP can carry data that will not be passed to TCP/UDP.



1.3 Note:

- An IP datagram has a total of 20 bytes of header (assuming no options). If the datagram carries a TCP segment, then each datagram carries a total of 40 bytes of header (20 bytes of IP header plus 20 bytes of TCP header) along with the application-layer message.

Data packet



1.4 IPv6:

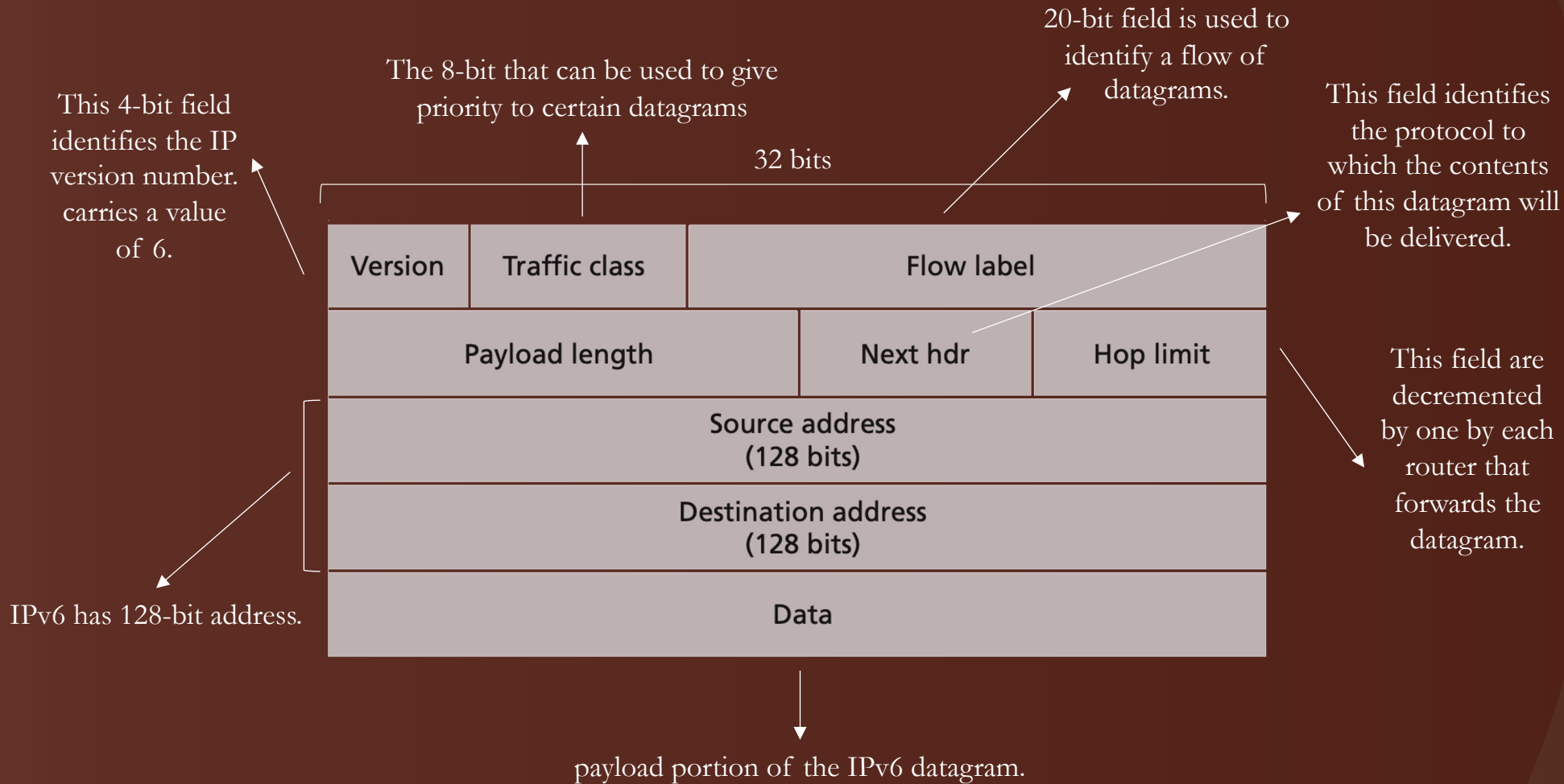
- The most important changes introduced in IPv6 are evident in the datagram format:
 - Expanded addressing capabilities.
 - A streamlined 40-byte header.
 - Flow labeling.
- IPv6 has introduced a new type of address, called an **anycast address**, that allows a datagram to be delivered to any one of a group of hosts.

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Site Prefix Subnet ID Interface ID

1.5 IPv6 Datagram format:

- The key fields in the IPv6 datagram:



1.6 IPv6 vs IPv4 :

○ Comparing the IPv6 datagram format with the IPv4 datagram format:

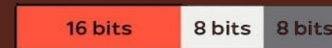
- Fragmentation/Reassembly.
- Header checksum.
- Options.

IPv4

Version	Header length	Type of service	Datagram length (bytes)	
16-bit Identifier			Flags	13-bit Fragmentation offset
Time-to-live	Upper-layer protocol		Header checksum	
32-bit Source IP address				
32-bit Destination IP address				
Options (if any)				
Data				

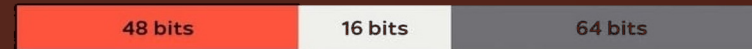
IPv4

5.57.226.105



IPv6

2a00:de40:10:313:0:0:0:105

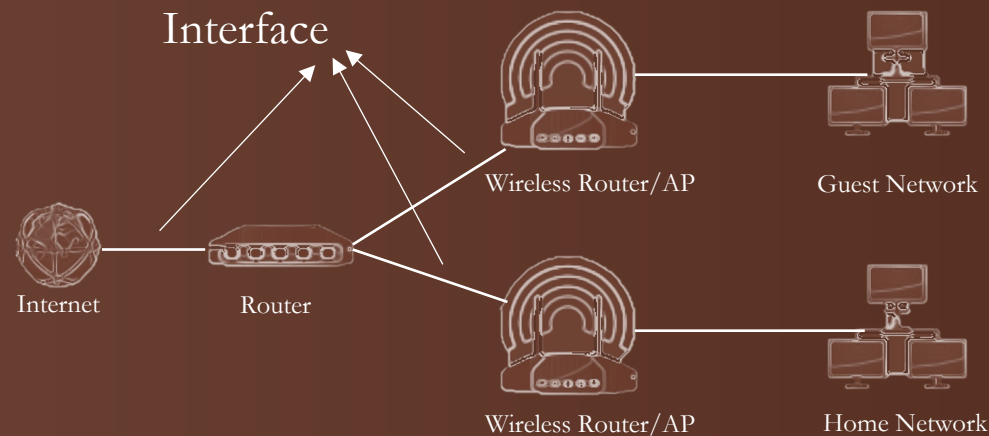


IPv6

Version	Traffic class	Flow label		
Payload length		Next hdr	Hop limit	
Source address (128 bits)				
Destination address (128 bits)				
Data				

2.1 Router's Interfaces:

- How hosts and routers are connected into the Internet?
- The boundary between the host and the physical link is called an **interface**. Because a router's job is to receive a datagram on one link and forward the datagram on some other link, a router necessarily has two or more links to which it is connected.
- The boundary between the router and any one of its links is also called an interface. A router thus has multiple interfaces, one for each of its links. IP requires each host and router interface to have its own IP address.

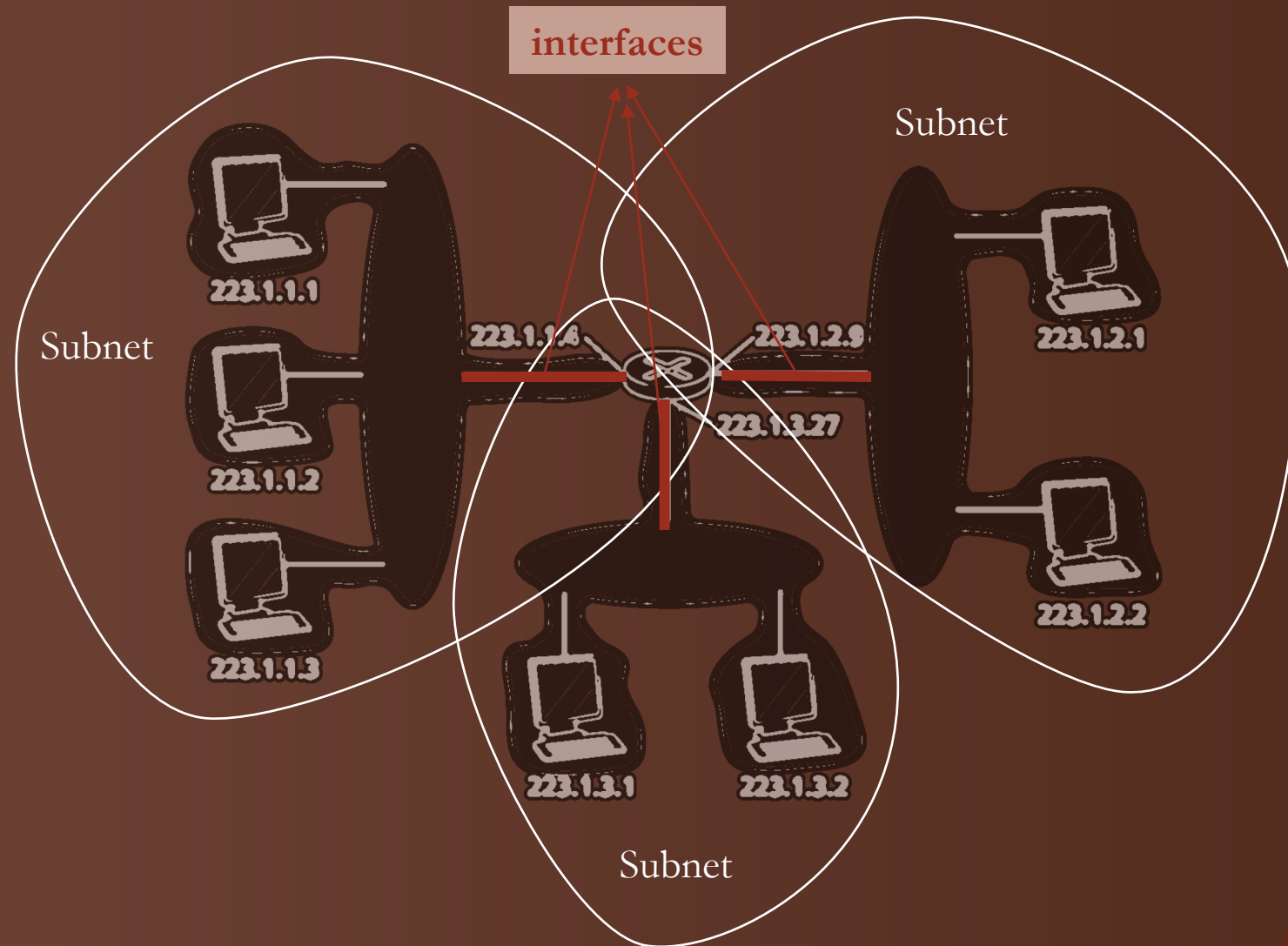


2.2 IPv4 Structure:

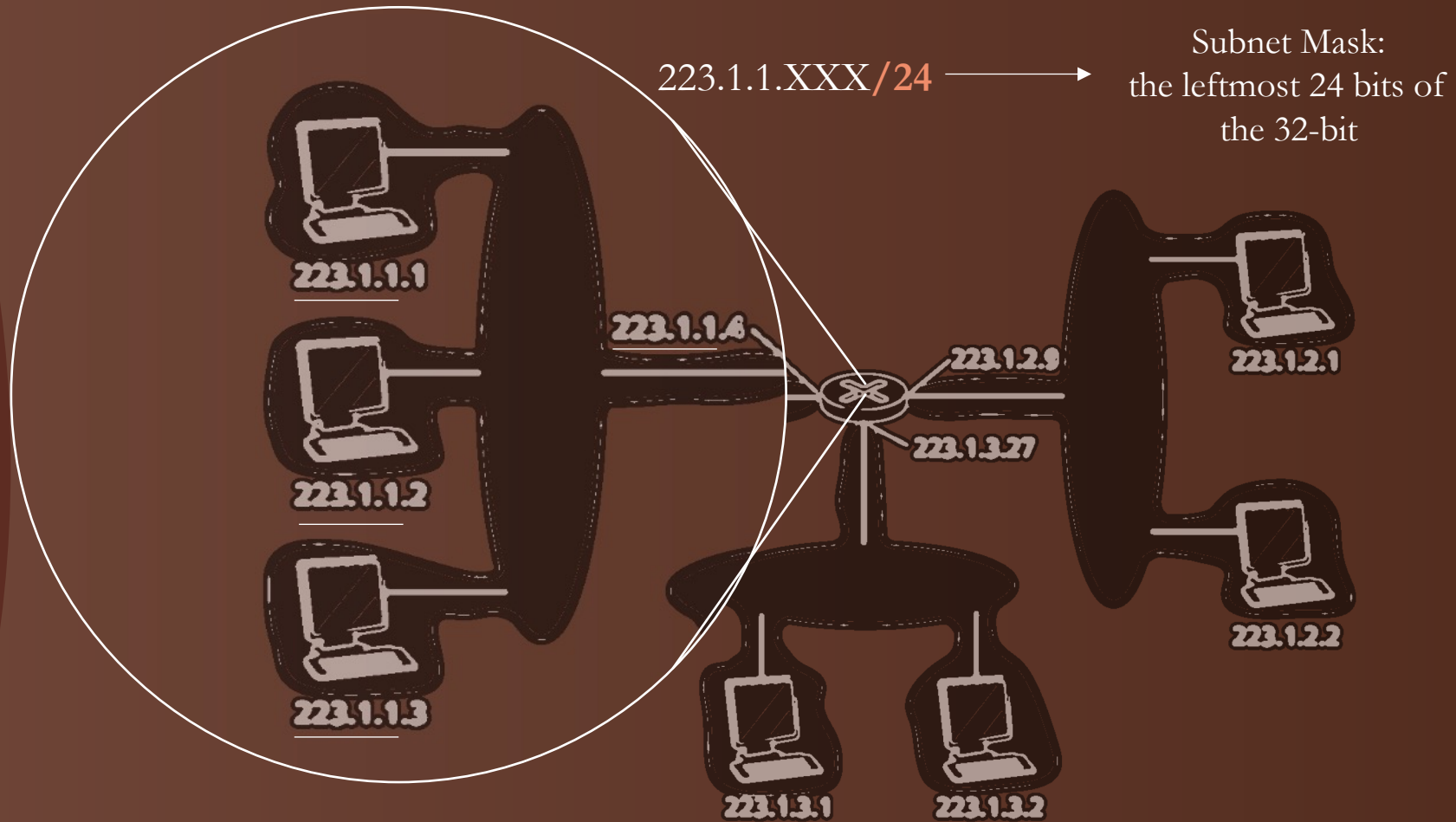
- Each IP address is 32 bits long (equivalently, 4 bytes), and there are thus a total of 2^{32} (or approximately 4 billion) possible IP addresses.
- These addresses are typically written in so-called **dotted-decimal notation**.
- Example:
 - consider the IP address 193.32.216.9;
 - The 193 is the decimal equivalent of the first 8 bits of the address;
 - The 32 is the decimal equivalent of the second 8 bits of the address, and so on.
 - Thus, the address 193.32.216.9 in binary notation is:

11000001 00100000 11011000 00001001

2.3 Subnet's IP:

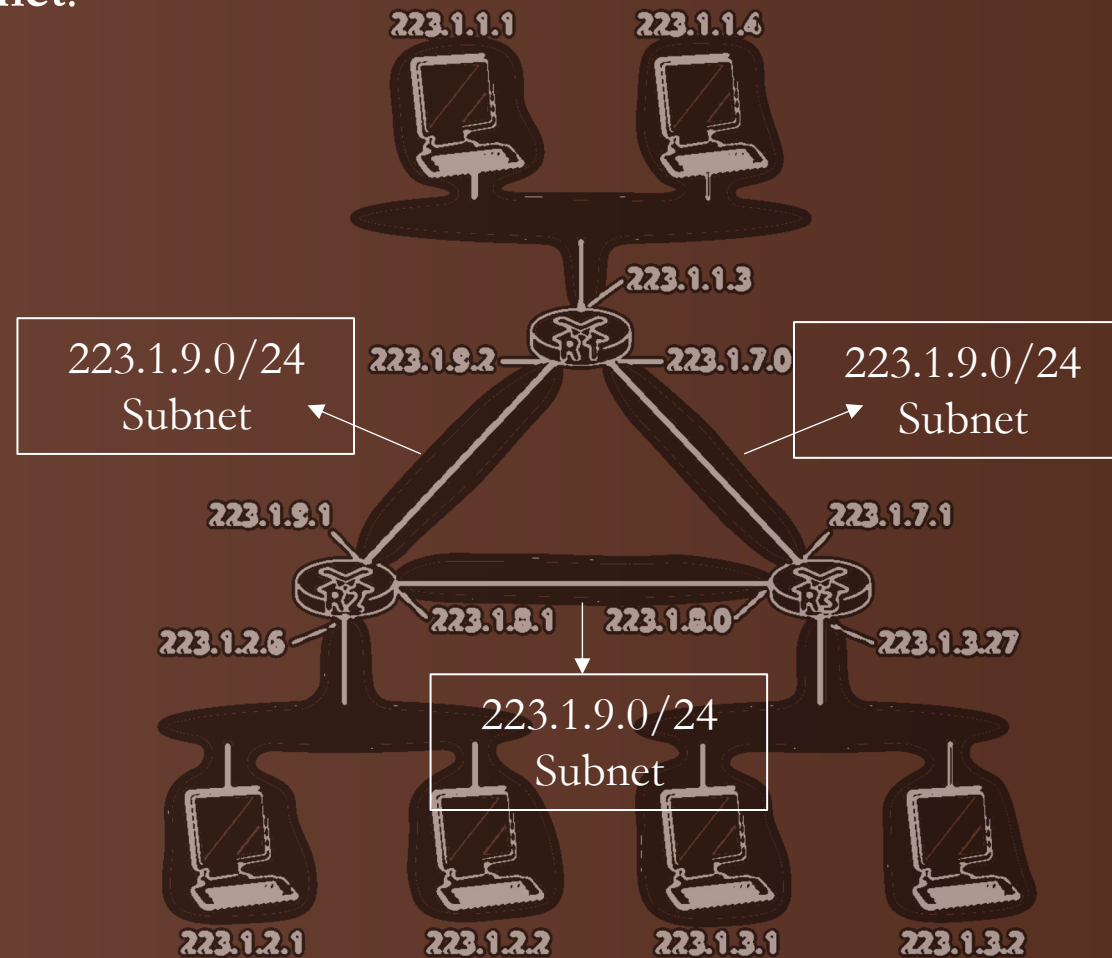


2.3 Subnet's IP:



2.3 Subnet's IP:

- To determine the subnets, detach each interface from its host or router, creating islands of isolated networks, with interfaces terminating the end points of the isolated networks. Each of these isolated networks is called a **subnet**.



2.4 CIDR:

- The Internet's address assignment strategy is known as Classless Interdomain Routing.
- CIDR generalizes the notion of subnet addressing. As with subnet addressing, the 32-bit IP address is divided into two parts.
 - x bits for *outside* of the organization's network.
 - 32-x bits for *within* the organization's network.
- This considerably reduces the size of the forwarding table in these routers.
- Classful Addressing:

8-bit	] subnet addresses are: <i>class</i>	[	A
16-bit			B
24-bit			C

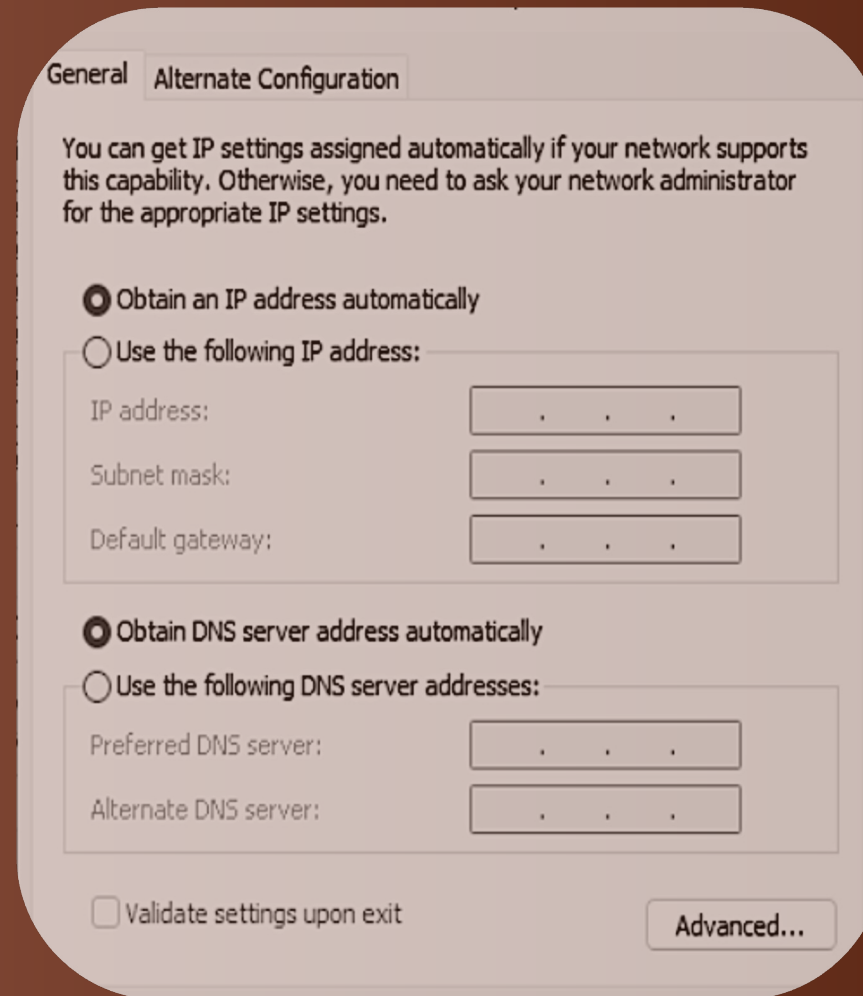
a.b.c.d/x

Network Prefix

3.1 DHCP:

- DHCP (Dynamic Host Configuration Protocol):

- There are 2 ways of assigning IP:
 1. Manually.
 2. By DHCP.



The image shows a network configuration window with two tabs: "General" and "Alternate Configuration". The "General" tab is selected. The window contains the following text and controls:

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address:

Subnet mask:

Default gateway:

☒ Obtain DNS server address automatically

☐ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

☐ Validate settings upon exit

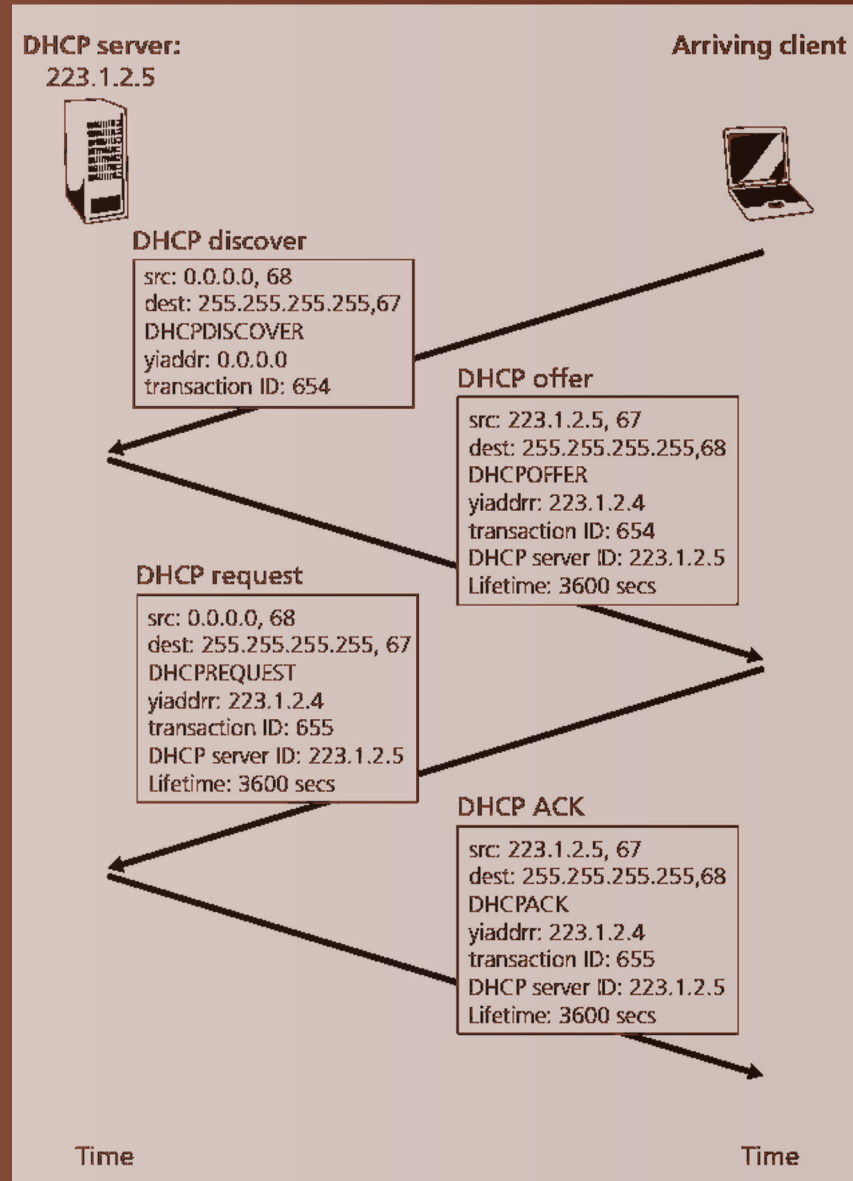
Advanced...

3.2 Lease & DORA:

- How much time can a client hold an IP address?
- The DHCP Lease time is the mechanism that allows DHCP servers to reclaim IP addresses, preventing a host from permanently retaining an IP address.
- DORA:

Discover	Client -> Server	Broadcast
Offer	Server -> Client	Broadcast/Unicast
Request	Client -> Server	Broadcast
ACK	Server -> Client	Broadcast/Unicast

3.3 DORA (Continuing):





The End.